

Technical Specifications
For
Auto Transfer Switch (ATS),
Phase Sequence Corrector (PSC)
&
AC Distribution Board (ACDB)
At
National Load Despatch Centre (NLDC),
Katwaria Sarai, New Delhi

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1. Introduction

The proposed AC Distribution Boards (ACDB) Panel, Automatic Transfer Switch (ATS) Panel, Phase Sequence Corrector (PSC) and other equipment envisaged under this contract shall be supplied, commissioned, maintained etc. at National Load Despatch Centre (NLDC), B-9, Qutab Institutional Area, Katwaria Sarai, New Delhi and shall be used to connect 2x 40 kVA UPS system and 250 A Raw Power from LT Panel to facilitate power supply for NLDC POSOCO premises.

2. Intent of the project

The intent of this Project is as follows:—

- Testing, Supply, Installation of new proposed ACDB's, ATS and PSC and associated equipment as per the specification and BoQ attached at **Annexure-I** and **Annexure-II** respectively.
- Warranty of all the systems supplied under this contract for 01 year.
- Extended warranty of the supplied system for further 01 year.

2.1. Automatic Transfer Switch

The Contractor shall provide two no's of Automatic Transfer Switch (ATS) of 160 A and 250 A as per the technical specification specified under this contract. ATS Panel of 160 A and 250 A shall be fed through dual incoming supply from LT panel. ATS Panel should be wall mounted. ATS shall distribute power to the Phase Sequence Corrector equipment and provide protection against failures on feeder circuits. The scope for ATS includes supply, testing, installation, wiring, cabling, and commissioning of all associated hardware's under this contract. The contractor shall supply and install MCCBs, connecting joints/cables and all associated items of adequate ratings for ATS Panel of 160 A and 250 A for providing input to the 150 A Phase Sequence Corrector (PSC) and 250 A Phase Sequence Corrector respectively (as per Annexure-I). ATS Panel should have manual change over provision so that in case of fault in ATS Panel, power supply may be manually bypassed and interruption may be avoided.

The contractor shall extend safety earth connections from ATS panel to nearest earth grid /riser or earth pit using either suitably sized Copper cable or through 50 X 6 sq. mm GI strips.

2.2. Phase Sequence Corrector

The Contractor shall provide two no's of Phase Sequence Corrector (PSC) Panel of 250 A and one Phase Sequence Corrector (PSC) Panel of 150 A as per the technical specification specified under this contract. PSC Panel of 150 A and 250 A shall be fed through incoming supply from ATS panel of 160 A and 250 A respectively. PSC Panel should be wall mounted. PSC Panel of 150 A shall be connected with appropriate size of copper cable from 160 A ATS. PSC Panel of 250 A shall be connected with appropriate size of copper cable from 250 A ATS. 250 A PSC Panel shall distribute power with appropriate size of flexible copper cable to the 250 A ACDB Output Panel. 150 A PSC Panel shall distribute power with appropriate size of copper cable to the terminal of 2x40 KVA UPS system. The scope for PSC includes supply, testing, installation,

wiring, cabling, and commissioning of all associated hardware's under this contract. The contractor shall supply and install MCCBs/contactors, Surge protection devices etc. and all associated items of adequate ratings for providing input to the 2x40 KVA UPS and 250 A Output ACDB (as per Annexure-I).

The contractor shall extend safety earth connections from PSC panel to nearest earth grid /riser or earth pit using either suitably sized Copper cable or through 50 X 6 sq. mm GI strips.

2.3. AC Distribution Board Requirements

The Contractor shall provide two no's of AC power distribution boards (ACDBs) of 250 A and 150 A as per the technical specification specified under this contract. The distribution boards shall distribute power to the Employer/ Owner equipment and provide protection against failures on feeder circuits. The scope for distribution boards includes supply, testing, installation, wiring, cabling, and commissioning of all associated hardware's under this contract. The contractor shall supply and install MCCBs, MCBs, Surge protection devices etc. and all associated items of adequate ratings for ACDB (as per **Annexure-I**).

A conceptual configuration showing the components and arrangements of 'ATS', 'PSC' and 'Output ACDB' is shown in **figure-1** and **figure-2**. The size and characteristics of each MCCB, MCB & Surge Protection Devices (SPDs), Automatic Transfer Switch (ATS) and Phase Sequence Corrector shall be determined and provided as specified under **Annexure-I**. The quantity of ATS, PSC, MCCBs, MCBs, Surge Protection Devices (SPDs), cables etc. shall be supplied as per **Annexure-II**. The MCCBs conforming to IEC-60947-2 and IS 13947-2/IEC 60947-2 shall be of four (04) Pole type of requisite rating. All the MCCBs shall provide over-current, short circuit protection and coordinate with associated breakers in upstream & downstream such that faults are cleared reliably with discrimination.

The contractor shall extend safety earth connections from ATS, PSC & Output ACDB panel to nearest earth grid /riser or earth pit using either suitably sized copper cable or through 50 X 6 sq. mm GI strips.

2.3.1. Output ACDB

The Contractor shall provide Output ACDB panel of 250 A rating and 150 A rating for further distribution of power to the different equipment. The rating shall be of minimum as stated in **Annexure-I, Hardware Requirements**. This distribution board shall be suitable for free-standing or wall mounted type which shall be finalised after the site survey by the contractor after the award of contract. In this ACDB, total number of MCCBs and MCBs shall be provided as per the BoQ.

One number of 4-pole Bus Coupler MCCB of requisite rating for each ACDB Panel shall be provided which shall be kept in normally closed (NC) condition when both UPS are running in parallel. This MCCB shall have suitable locking mechanism to avoid inadvertent opening while UPS are running in parallel.

250 A ACDB Output Panel shall be connected with 250 A ATS Panel with one 70 sq.mm. 4 Core Flexible cable and other 185 sq.mm. aluminium armoured cable. 70 sq.mm., 4 Core Flexible Copper cable shall be supplied and laid at the site as per BoQ.

150 A ACDB Output Panel shall be connected with 160 A ATS Panel with suitable two 25sq.mm. flexible copper cable. 25 sq.mm., 4 Core Flexible Copper cable shall be supplied and laid at the site as per BoQ.

All the MCBs supplied under the project shall be of Curve C current characteristics to have faster fault clearing time and shall conform to IEC-60898 and IS 8828.

Output feeders shall have indicating lamps for each phase & both section of bus charged indication. Two no's of digital Voltmeter with selector switch for all phases on section of each bus shall be provided.

2.3.2. Enclosures/Panels

The equipment of ACDB shall be physically mounted either in a free standing or in a wall mounted enclosure/panel as finalized after the site survey by the contractor after the award of contract. Approval must be taken for the size of the ACDB, ATS and PSC panels after the site survey by the contractor after the award of contract. The MCCB and sub-assemblies shall be easily replaceable and maintainable. Cable entry shall be from the bottom/top of the enclosures (to be finalized after the site survey by the contractor after the award of contract). The Contractor shall state the type, size, and weight of all enclosures and indicate the proposed manner of installation. The applicable degree of protection of enclosures shall be at least IP54. Each ACDB and equipment within ACDB enclosures shall be clearly labelled to identify the enclosure/equipment.

3. Inspection and Testing

All equipment supplied under the project shall be factory tested before despatch to site as per requirement specified under this section. General inspection & testing requirements are as follows:

3.1. Factory Test

A factory test shall be conducted on all the offered equipment which shall include, but not be limited to the following, appropriate to the equipment being tested:

- a) Verification of all functional characteristics and requirements specified
- b) Inspection and verification of all construction, wiring, labelling, documentation and completeness of the hardware.

Before the start of factory testing, the Contractor shall verify that all changes applicable to the equipment have been implemented. As a part of the factory tests, unstructured testing shall be performed to allow Employer/ Owner representatives to verify proper operation of the equipment under conditions not specifically tested in the above structured performance test. A minimum of 8 hours of the factory test period shall be reserved for unstructured testing. The Contractor's test representative shall be present and the Contractor's technical

staff members shall be available for consultation with Employer/ Owner personnel during unstructured test periods. All special test facilities used during the structured performance test shall be made available for Employer/ Owner's use during unstructured testing. In addition to above, a list of factory tests specified at the end of the respective sections of this equipment shall also be performed.

3.2. Field Performance Test

After the equipment has been installed at site, the Contractor shall start up and check the performance of the equipment at field location. All hardware shall be aligned and adjusted, interfaces to all inputs and outputs installed, operation verified, and all test readings recorded in accordance with the Contractor's recommended procedures. The field performance test shall exhibit all functions of the equipment and duplicate factory test. All variances must be corrected prior to the start of the field performance test.

Contractor shall submit the procedure for Factory and Site test and take approval from the Owner.

4. Documentation

Contractor shall visit the site before submission of any documents. After the discussion and visit of site only contractor shall submit the drawing of panels for approval from the owner. Drawing shall also indicate the arrangements of all the equipment envisaged under each panels. Data requirement sheets (DRS) for equipment and preferred make are enclosed at Annexure-I and Annexure-III respectively, however, it shall be finalised in consultation with owner before start of work. These will be duly filled and complied by the bidder & details thereof submitted along with the bid.

5. Bid evaluation or Price

- a) List of equipment are mentioned in BoQ, however, if it has been felt that some additional items shall be required to meet the entire scope of work, same may be intimated while submitting bidding documents with price. After the award of LoA, if any additional items are required, Owner shall not pay any amount on that head.
- b) The Contractor shall quote for the Supply, Commissioning, Warranty, extended warranty and Taxes charges separately.

5.1. Work Schedule

Two (02) month from date of issuance of Letter of Award for supply, testing and commissioning of proposed system.

5.2. Defect Liability Period (Warranty)

The defect liability period is for 12 months, the period shall commence after issuance of TOC by the Engineer-in-charge or his/her authorized representative. During this period the vendor shall be responsible for any repairs or replacement of any defective part and shall rectify the installation free-of-cost to the owner.

Contractor shall provide separate price for extended warranty, initially it is envisaged for one year (after the completion of defect liability period) whereas is desired owner reserves the right to extend it for further 3 years at same price and terms & condition.

5.3. Payment Terms

- a) **For Supply, Testing and Commissioning:** 90% payment shall be made after successful commissioning and testing of the system and after verification by Engineer-in-charge.
- b) **For Warranty:** 10% payment shall be released after lapse of defect liability period and after verification by Engineer-in-charge.
- c) **Extended Warranty:** Payment shall be released on six monthly basis and after the lapse of defined period.
- d) Tax shall be deducted at source as per the statutory laws.

5.4. Penalty Clause

System shall be supplied, tested and commissioned within two (02) month of acceptance of Letter of Award (LoA). Delay thereafter shall be considered to levy penalty as specified under the contract.

5.5. Cancellation of Contract in Full or Part

If vendor fails to deliver the services as per the scope of the work detailed in this LoA, due to any lapse from the vendor side, POSOCO reserves the right to cancel the order (including extended warranty period) within one month of notice. In the case of termination / cancellation of the order, the vendor shall not be entitled to recover from POSOCO any amount by way of damages, loss or otherwise. However, in case of cancellation of contract during extended warranty, pro-rata amount will be released.

6. General Requirements

The Contractor is encouraged to offer standard products and designs. However, the Contractor must conform to the requirements and provide any special or additional equipment/s or software necessary to meet the requirements stated in the specification.

The employer reserves the right of execution of works within the stipulated quantity variation of **20%** provision other than those indicated in the annexures at the same rates, terms and conditions. The Bidder's proposal shall address all functional and performance requirements within this specification and shall include sufficient information and supporting documentation in order to determine compliance with this specification without further necessity for enquiries.

An analysis of the functional and performance requirements of this specification and/or site surveys, design, and engineering may lead the Contractor to conclude that additional items and services are required that are not specifically mentioned in this specification. The

Contractor shall be responsible for providing at no added cost to the Employer, such materials and services shall be considered to be within the scope of the contract.

The bidders are advised to visit site (at their own expense), prior to the submission of proposal, and make surveys and assessments as deemed necessary for proposal submission. The successful bidder (Contractor) is required to visit sites. The site visits/routes shall include all necessary surveys to allow the contractor to perform the design and implementation functions. The Contractor shall inform their site survey schedule to the Employer/Owner well in advance.

Successful bidder shall visit the site where this equipment shall be supplied, commissioned and maintained.

7. Survey of site

The contractor shall conduct a survey of the NLDC premises and identify the readiness of site with respect to access, installation, requirement of conduits and raceways, identification of routes for laying cables, finalizing the layout and placement of the ACDB, ATS and PSC verifying special requirements with respect to earthing and air-conditioning. ***After the survey of site size of panels*** (however, it is desired that all the components shall be considered as copper based in order to reduce the size of panels due to space constraints), ***cable entry etc shall be decided and accordingly drawing shall be submitted by the contractor.***

Annexure - I

Hardware Specifications

a) ATS (Automatic Transfer Switch)

S. No.	Description	Minimum Required Features	Bidder Compliance
1.	Ampere Capacity	250 A, 4P ATS & 160 A, 4P ATS	
2.	Miscellaneous	1. Phase Indicating Lights set R-Y-B-22.5 mm dia; LED Type. 2. Phase Indicating Lights in Incoming and Output of ATS 3. Indicating Lights for On/Off/Trip 4. Manual Change Over Provision 5. 6A SP Control MCB 10 KA C-Curve 6. Voltmeter-96x96 mm.sq., CL-1 Analog, 0-500V 7. Voltmeter Selector Switch 3P+Off; 6A	

b) Phase Sequence Corrector

S. No.	Description	Minimum Required Features	Bidder Compliance
1.	Ampere Capacity	250 A & 150 A , 3P	
2.	Applicable standards	IEC 60947-2	
3.	Miscellaneous	1. Spreader Terminals, Auxiliary Contact, LED indicator for Power ON, Power Terminal Blocks (as per requirements) etc. 2. Phase Indicating Lights set R-Y-B- 22.5 mm dia; LED Type	

c) MCCB: Bus Coupler and Outgoing Feeders

S. No.	Description	Minimum Required Features	Bidder Compliance
1.	Applicable standards	IEC 60947-2	
2.	Characteristics	250 A, 4P, 25 kA with adjustable thermal O/L (80-100 & fixed O/L & S/C release) + Door Interlocked	
3.	Ampere Capacity	250 A ,4P	
4.	Miscellaneous	Rotary Handle, Spreader Terminals, Auxiliary Contact, LED indicator for Power ON, Power Terminal Blocks (as per requirements) etc.	

d) MCCB: Incoming Feeders

S. No.	Description	Minimum Required Features	Bidder Compliance
1.	Applicable standards	IEC 60947-2	
2.	Characteristics	100 A, 250 A, 4P, 25 kA with adjustable thermal O/L (80-100 & fixed O/L & S/C release) + Door Interlocked	
3.	Indication Lamps:	1. Phase Indicating Lights set R-Y-B-22.5 mm dia; LED Type: 230 V AC. 2. Indicating Lights for On/Off/Trip: 220 V AC. 3. 6A SP Control MCB 10 KA C-Curve	
4.	Ampere Capacity	250 A, 4P ; 100 A ,4P	
5.	Voltmeter	1. Voltmeter-96x96 mm.sq., CL-1 Analog, 0-500V 2. Voltmeter Selector Switch 3P+Off; 6A	
6.	Miscellaneous	Rotary Handle, Spreader Terminals, Auxiliary Contact, Surge Protection Device for class B&C, Power Terminal Blocks (as per requirements) etc.	

e) MCB:

S. No.	Description of the Features	Minimum Required Features	Bidder Compliance
1.	Applicable standards	IEC 60947-2	
2.	Characteristics	32A-DP, 63A-4P, 4-pole, 10 kA, 'C' Curve	
3.	Ampere Capacity	32 A, 63 A	
4.	Miscellaneous	Power Terminal Blocks (as per requirements)	

f) Main Bus Bars:

Sl. No.	Description of the Features	Minimum Required Features	Bidder Compliance
1.	Material Size:	Copper: Phase-1x40x6 or Eq. Cross Section Neutral-1x40x6 or Eq. Cross Section	
2.	Sleeve	Heat shrinkable Black PVC sleeve with color coding at regular interval for phase and neutral	
3.	Earth Busbar	1x25x3 Cu	

g) Copper Cable:

Sl. No.	Description of the Features	Minimum Required Features	Bidder Compliance
1.	Material Type and Size:	Copper: 4 Core Flexible Cables upto 1.1kv grade as per IS:694 Sizes :25 and 70 sq. mm	
2.	Options	- Stranded / annealed Copper Insulation-HFFR Insulated industrial cables 1100V	

h) Panels (ATS, Phase Sequence Corrector, Input and Output)

Sl. No.	Description of the Features	Minimum Required Features	Bidder Compliance
1.	Type of Product	Indoor, Floor/Wall Mounting, Cubicle Fixed Type, Double Front	
2.	Supply voltage	415 +/- 10%, 3-phase 4-wire, 50 Hz +/- 5%	
3.	The thickness of the structural frames and load bearing members	For load bearing 2 mm and others 1.6 mm	
4.	Degree of Protection as per IS:13947 for enclosure only	IP-54	
5.	Fabrication: a. Enclosure frame b. Doors c. Front Side, partition d. Gland Plate e. Base Channel	a. 2 mm CRCA sheet steel b. 1.6 mm CRCA sheet steel c. 1.6 mm CRCA sheet steel d. 3 mm sheet steel e. 2.5 mm CRCA Base Frame (ISMC-75 Black)	
6.	Cable Entry	Bottom/Top	
7.	Paint/Construction: a. Pre-treatment b. Paint Type c. Paint Shade (Exterior and Interior) d. Wiring e. Short Circuit Level	a. 9 tank process b. Powder Coated c. RAL 7032 Smoke Grey, as per if 5 & coated with Peelable Compound finish. d. 1100 V grade PVC insulated wire of 1.5 Sq.mm for control and 2.5 sq. mm for CT circuit. e. 20 kA/ 1 sec.	
8.	Min. Clearance between (for Horizontal and Vertical bus bar)	1. Phase to phase: 25 mm 2. Phase to neutral: 25 mm 3. Phase to Earth: 25 mm 4. Neutral to Earth: 25 mm	
9.	Incoming/Outgoing feeders	As per SLD	
10.	Applicable standards	IS 8623 (Part-I) 1993	
11.	Name plate	It shall be non-rusting metal or 3 ply lamicaid with white engraved lettering on black back ground	
12.	Others	1. Partition between compartment & cable alley shall be non-metallic part. 2. All metallic case like relays, instruments & other equipment shall be connected to earth with 2.5 sq.mm Cu Wire, Green Colour. 3. SMC support shall be 450 mm between centre to centre.	

Annexure-II

Bill of Quantity (BoQ)

List of items, quantities and services assessed for Project

S. No.	Description	Quantity
Supply (including		
1.	Automatic Transfer Switch (ATS) Panel (160A)	01 set
2.	Automatic Transfer Switch (ATS) Panel (250A)	01 set
3.	Phase Sequence Corrector Panel (150 A)	01 set
4.	Phase Sequence Corrector Panel (250 A)	02 set
5.	Outgoing ACDB Panel (including 2 nos. incomer MCCB of 250 A and 1 nos. bus coupler MCCB of 250 A, outgoing 12 nos. MCB of 63 A and 18 nos. MCB of 32 A)	01 set
6.	Outgoing ACDB Panel (including 2 nos. incomer MCCB of 150 A and 1 nos. bus coupler MCCB of 150 A, outgoing 8 nos. MCB of 63 A and 6 nos. MCB of 32 A)	01 set
7.	25 Sq.mm , 4 Core , Flexible Copper Cable	90 metre
8.	70 Sq.mm , 4 Core , Flexible Copper Cable	140 metre
Extended Warranty		
9.	One (01) year Warranty Period	
Transportation		
10.	Transportation of equipment	
Installation and commissioning		
11.	Installation and commissioning (including internal wiring, etc.)	

Note:

- In above mentioned bill of quantity (BoQ), quantity mentioned with unit "Set" includes all the corresponding items mentioned under hardware specification (annexure-I). No additional cost on that head shall be reimbursed by owner.*
- Costs of all internal wiring within the panel are also inclusive under above quoted price. No additional cost on that head shall be reimbursed by owner.*
- Initially extended warranty envisaged is for one year, however, owner reserves the right to extend it for further 3 years at same price and terms & condition.*

Annexure-III

List of Preferred Make

Sl. No.	Description	Preferred Make
1.	Automatic Transfer Switch (ATS)	Schneider/Havells/Siemens/General Electric (GE)
2.	Phase Sequence Corrector	Schneider/Havells/Siemens/Indokopp/C&S/MDS/C&S/GE/L&T
3.	MCB	Schneider/Havells/Indokopp/C&S/MDS/C&S/GE/L&T
4.	MCCB	Schneider/C&S/GE/ L&T
5.	MCCB (for LT Panels)	Schneider Electric
6.	Surge Protection Devices	Schneider/Phoenix/JVM-Bradley/Batterman
7.	Indicating Lights LED	Schneider/ C&S/Vaishno/Esbee/Teknic
8.	Voltameter	AE/Rishabh
9.	Voltameter selector switch	Switron/Salzer/Kaycee
10.	Rotary Handle	Schneider
11.	Spreader Terminals	FTC/L&T
12.	Auxiliary Contact	Schneider
13.	Power Terminal Blocks	Elmex/Connectwell/Phoenix
14.	Cables	Reputed manufacturer

Figure

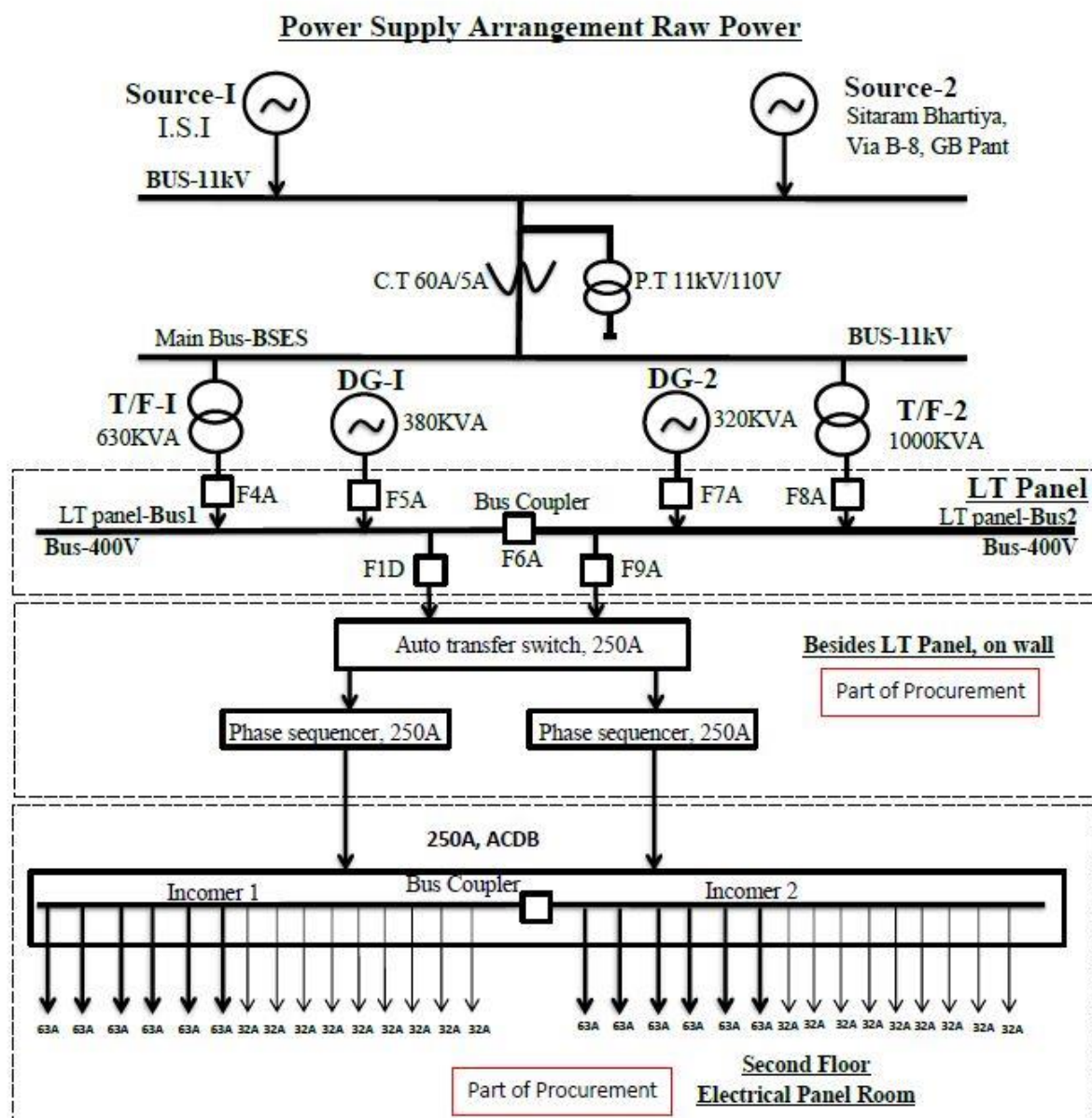


Figure 1: - Raw Power Supply Arrangement

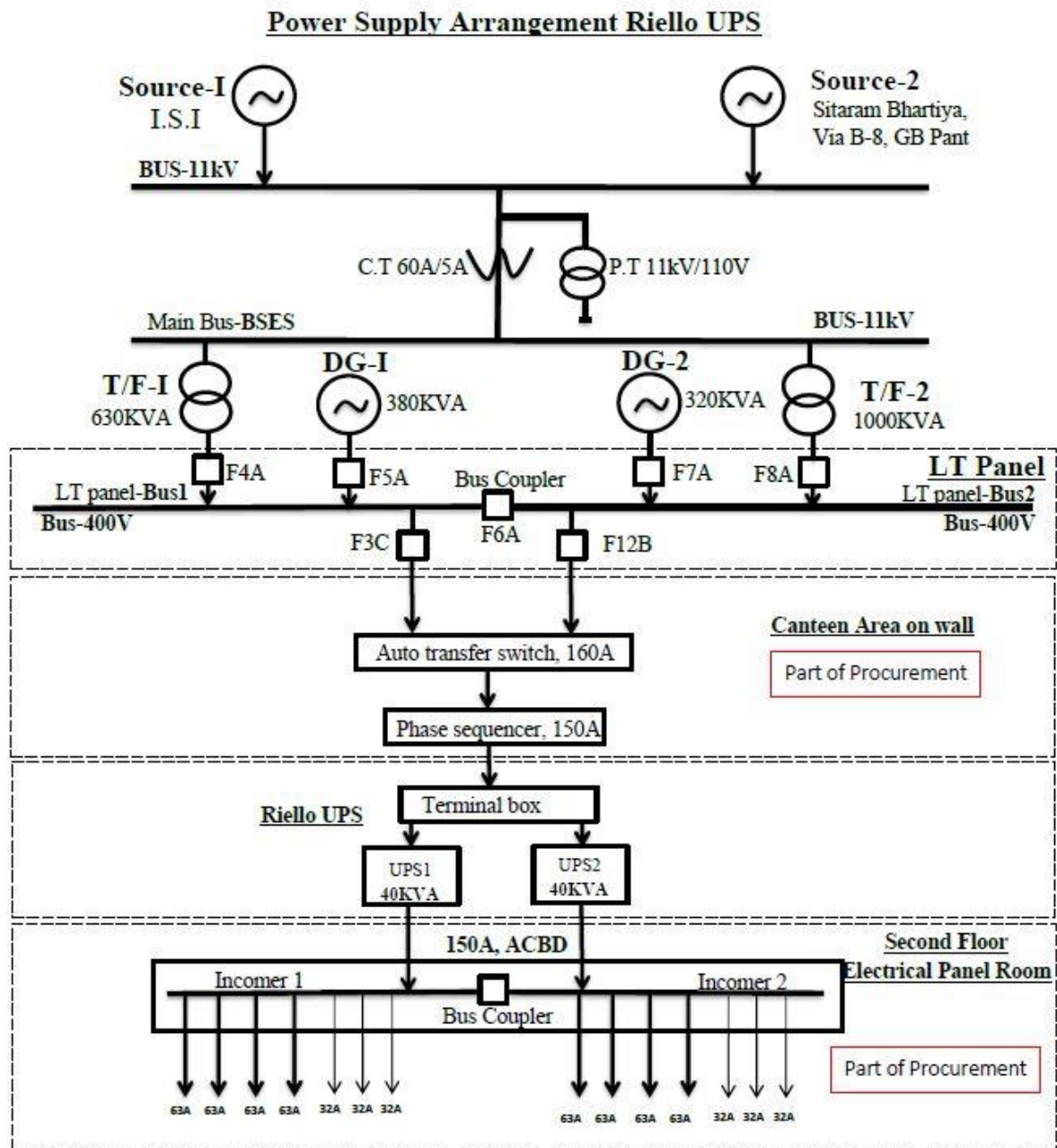


Figure 2: - UPS Power Supply Arrangement